

EXECUTIVE BRIEF

Maintain First, Replace With Evidence

What leaders should authorize, require, and refuse

Production Unit	PU-B07-C56
Decision status	Controlled learning asset
Audience	Executives, programme owners, statisticians, technical leads
Data	Synthetic examples only

Executive takeaway: Prioritize infrastructure maintenance and replacement using condition, service criticality, failure probability, consequence, lifecycle cost, uncertainty, equity, and accountable asset authority.

Why it matters

Replacing the most visible asset first can waste resources while less visible critical failures remain unaddressed.

Illustrative application

A municipality compares preventive maintenance, rehabilitation, and replacement scenarios for a deteriorating water asset.

Minimum approval conditions

- Named accountable owner and lawful purpose
- Defined evidence, provenance, reference period, population, and geography
- Decision-relevant validation and visible uncertainty
- Documented abstention, referral, override, and incident processes
- Accessibility, equity, security, retention, and retirement review

Non-negotiable boundary

The analytical system may inform review. It may not independently authorize benefits, sanctions, enforcement, clinical treatment, school exclusion, displacement, or another consequential action.